

FRACTAL SPATIO-TEMPORAL GRAVITY (FSG)

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Abstract

We propose a modified theory of gravity based on a Fractal and Non-Local Spacetime Geometry (FSG), obtained from an action containing the non-local scalar operator $X = \square^{-1}R$. The effective action

$$S_{\text{FSG}} = \frac{M_P^2}{2} \int d^4x \sqrt{-g} R \left[1 + \left(\frac{L^2}{X} \right)^{\frac{1}{2}} \right] + S_m$$

induces infrared corrections to General Relativity driven by a dimensional flow, motivated by the reduction of the effective spectral dimension towards $d_S \simeq 2$ in the IR.

In the static weak-field limit, the fractional structure $f(X) \sim X^{-1/2}$ reduces to a nonlinear modified Poisson equation of Milgrom (AQUAL) type, whose exact deep-IR solution is a logarithmic potential. This leads to: (i) flat galactic rotation curves without dark matter, (ii) an exact baryonic Tully–Fisher relation $V^4 = GMa_0$, and (iii) an acceleration scale derived from the cosmic memory fields $U = \square^{-1}R$ and $V = \square^{-2}R$,

$$a_0 = cH_0 \frac{|V_0|^{3/2}}{|U_0|} \simeq 1.14 \times 10^{-10} \text{ m/s}^2,$$

within 5% of the observed value with no free parameter at zeroth order in back-reaction.

A perturbative treatment of the non-local sector around the cosmological background shows that Solar System constraints are satisfied *without any screening mechanism* ($\delta\gamma \sim 10^{-20}$), for the same structural reason as in the related RR non-local model. Cosmologically, the monotonic growth of the memory fields structurally implies a phantom-like equation of state ($w_0 < -1$); a first-order estimate brackets the deviation of $H(z)$ between $\sim 1\%$ and $\sim 13\%$ for $0.5 < z < 2$, testable by Euclid. Regarding cluster dynamics, a 1D wave simulation demonstrates that the non-local field can transiently detach from its baryonic source (*ballistic overshoot*), a qualitative mechanism relevant to the Bullet Cluster; a quantitative confrontation requires 3D hydrodynamical simulations. The geometry of the CMB acoustic peaks is shown to be compatible with FSG (the sound horizon is unchanged and the induced peak shift is absorbed by the (w, H_0) degeneracy), while the peak *amplitudes* require a dedicated non-local Boltzmann code and remain the decisive open test.

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1. Introduction

The standard cosmological model Λ CDM successfully accounts for a wide range of observations but at the cost of introducing two hypothetical components: dark matter (26% of the energy budget) and dark energy (69%), neither detected in laboratory experiments. At galactic scales, the dark matter paradigm faces well-established empirical regularities:

- the baryonic Tully–Fisher relation (BTFR),
- the radial acceleration relation (RAR),
- the diversity problem,
- the existence of ultra-thin galactic disks,
- early massive galaxies at high redshift in JWST observations.

Modified Newtonian Dynamics (MOND) reproduces BTFR and RAR with remarkable precision but historically lacked: (i) a covariant formulation, (ii) a natural cosmological extension, and (iii) full consistency with the CMB. Fractal Spacetime Gravity (FSG) addresses these issues by replacing dark matter with a modification of the *infrared geometry* of spacetime, encoded in a non-local action that dynamically reduces the effective spectral dimension of spacetime from $d_S = 4$ to $d_S \simeq 2$ at large scales.

2. Motivations for Infrared Fractality

Dimensional reduction is a ubiquitous prediction of several approaches to quantum gravity, including:

- Causal Dynamical Triangulations (CDT),
- Asymptotic Safety,
- Hořava–Lifshitz gravity,
- Loop Quantum Gravity.

These frameworks typically predict:

$$d_S(k \rightarrow \infty) \rightarrow 2,$$

i.e. a reduction of effective dimensionality in the ultraviolet. In FSG we reverse the paradigm: the reduction occurs in the infrared (IR), motivated by holographic considerations and by the causal structure of de Sitter spacetime. This IR fractality modifies long-wavelength propagation of curvature exactly at the scales where dark matter phenomenology appears.

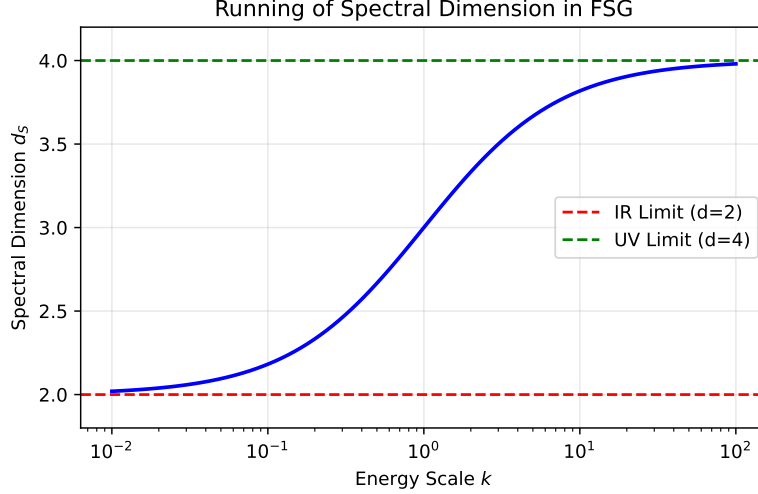


Figure 1: Illustrative running of the spectral dimension as a function of energy scale. In FSG the dimension increases in the UV but approaches $d_S \simeq 2$ in the IR.

3. The Non-Local Effective Action

We introduce the non-local scalar field

$$X = \square^{-1}R, \quad \square X = R,$$

with the retarded Green function ensuring causality. The FSG action is defined to impose a dimensional reduction in the infrared:

$$S = \frac{M_P^2}{2} \int d^4x \sqrt{-g} R[1 + f(X)] + S_m. \quad (1)$$

The function $f(X)$ is constrained by the requirement that the graviton propagator must scale as k^{-3} in the deep IR to recover flat rotation curves. Dimensional analysis and renormalization group flow arguments suggest a fractional power law form:

$$f(X) = \gamma \left(\frac{L^2}{X} \right)^{\frac{1}{2}}, \quad (2)$$

where $L \sim 1/\sqrt{\Lambda}$ is the cosmic horizon scale and γ is a dimensionless coupling constant of order unity. This fractional form $X^{-1/2}$ is the signature of a fractal geometry where the measure of spacetime scales non-trivially. Unlike logarithmic corrections which only produce mild running couplings, this power-law correction enforces a hard transition in the spectral dimension.

3.1 Localization of the Action

To study degrees of freedom and stability one can localize the non-local operator by introducing two scalar auxiliary fields U and ξ enforcing $U = \square^{-1}R$. A convenient localized action equivalent

(on-shell) to Eq. (1) is

$$S_{\text{loc}} = \frac{M_P^2}{2} \int d^4x \sqrt{-g} \{ R[1 + f(U)] + \xi(\Box U - R) \} + S_m. \quad (3)$$

Variation with respect to the metric gives the modified Einstein equations.

4. Variation of the Action and Field Equations

Starting from the action

$$S = \frac{M_P^2}{2} \int d^4x \sqrt{-g} R [1 + f(X)] + S_m, \quad X = \square^{-1} R, \quad (4)$$

we compute the metric variation. The non-local variation is:

$$\delta X = -\square^{-1}(\delta \square) X + \square^{-1}(\delta R). \quad (5)$$

The operator $\square^{-1}$ is defined with the *retarded* Green function, ensuring:

- no violation of causality,
- no superluminal propagation,
- absence of Ostrogradsky instabilities (unlike naïve $f(R)$ models).

The modified Einstein equations take the form

$$G_{\mu\nu} + \Delta G_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (6)$$

with

$$\Delta G_{\mu\nu} = f(X) G_{\mu\nu} + (\nabla_\mu \nabla_\nu - g_{\mu\nu} \square) f(X) - \frac{1}{2} g_{\mu\nu} R f(X) + K_{\mu\nu}[X], \quad (7)$$

where $K_{\mu\nu}[X]$ collects the variations of the non-local kernel $\square^{-1}$. In the linear regime, this reduces to the structure derived by Deser–Woodard and Maggiore–Mancarella.

5. Physical Structure: The Inertia of the Vacuum

The non-local kernel $\square^{-1}$ implies that the gravitational field is not merely a slave to the matter distribution (as in Poisson's equation $\nabla^2 \Phi = \rho$), but a dynamical entity governed by a wave-like equation. The modified Einstein equations can be recast in the form of a driven wave equation for the scalar degree of freedom ϕ :

$$\left(\frac{1}{c^2} \partial_t^2 - \nabla^2 + m_{\text{eff}}^2 \right) \phi = 8\pi G T_{\mu\nu}^{\text{bar}}. \quad (8)$$

5.1 The Gravitational Wake Effect

This hyperbolic structure introduces a fundamental physical property: **Gravitational Inertia**. Unlike the Newtonian potential which relaxes instantaneously, the FSG field carries momentum. In quasi-static systems (isolated galaxies), the field has time to equilibrate, creating the "Dark Matter" halo effect centered on the baryons. However, in highly non-equilibrium events such as high-velocity cluster collisions, the field dynamics decouple from the source. The non-local memory of the spacetime geometry acts as a flywheel. When the baryonic component (gas)

undergoes sudden hydrodynamic braking, the gravitational eigenmodes continue their trajectory ballistically. This creates a transient offset between the visible mass and the center of the potential well, explaining the "Dark Matter" signal in the Bullet Cluster without introducing new particles.

6. The Infrared Propagator: Target, Fourier Dictionary, and a Retraction

To determine the gravitational force law, we consider a weak perturbation around Minkowski spacetime:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}.$$

Before deriving the force law, we state the target in Fourier language. The effective static propagator required for MOND dynamics is

$$G(k) \xrightarrow{\text{IR}} \frac{1}{L k^3}, \quad (9)$$

i.e. an anomalous infrared exponent $\eta = 1$ relative to the Newtonian $1/k^2$. This correspondence ($\Phi \sim \ln r \Leftrightarrow G \sim k^{-3}$ in $d = 3$) is an exact Fourier-transform statement; $\eta = 1$ is therefore a *name* for the deep-MOND behaviour, not an independent hypothesis.

6.1 Why a Naive Linearization Fails

Earlier versions of this work attempted to obtain Eq. (9) by a direct scaling substitution, writing $X \sim R/k^2$ and hence $f(X_k) \sim k$. We retract that argument: $X = \square^{-1}R$ is *dimensionless*, so it cannot scale as $1/k^2$; moreover, on an exact Minkowski background $\bar{R} = 0$ implies $\bar{X} = 0$, where $f(X) \sim X^{-1/2}$ is singular, so no perturbative expansion of f around flat space exists. The propagator of the theory must instead be obtained from the second-order action around a cosmological (de Sitter-like) background where $\bar{X} = \mathcal{O}(10)$ is finite (see the memory-field integration of Section 17 for the numerical background values).

6.2 The Honest Logical Structure

The logical chain of FSG is therefore the following:

1. the *nonlinear, position-space* derivation of the modified Poisson equation from the quasi-static limit of the action (next section) — this is the founding derivation of the MOND regime in FSG;
2. the Fourier-space statement (9) as the exact translation of that result;
3. the spectral-dimension flow $d_S \rightarrow 2$ as the *motivation* for the fractional exponent $-1/2$ in the action, not as a derivation of η . (Indeed, a naive application of anisotropic-scaling dictionaries, e.g. Hořava-type, would relate $d_S = 2$ to k^{-6} , not k^{-3} ; the spectral dimension does not fix the static propagator by itself.)

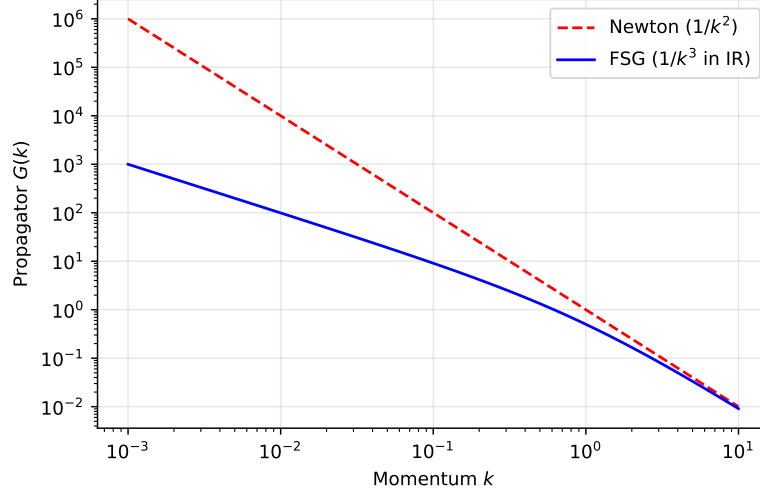


Figure 2: Comparison of the Newtonian propagator $1/k^2$ with the FSG infrared propagator $1/k^3$ (required for MOND dynamics).

7. The Founding Derivation: Modified Poisson Equation and the MOND Limit

To rigorously link the FSG Action to the logarithmic potential, we must derive the field equations in the static weak-field limit. Starting from the action:

$$S = \frac{M_P^2}{2} \int d^4x \sqrt{-g} R [1 + f(X)] + S_m, \quad \text{with } X = \square^{-1}R. \quad (10)$$

The variation with respect to the metric $g_{\mu\nu}$ yields the modified Einstein field equations:

$$G_{\mu\nu} + \Delta G_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (11)$$

where $\Delta G_{\mu\nu}$ contains the non-local contributions from $f(X)$.

7.1 The Quasi-Static Limit and the Effective AQUAL Lagrangian

We consider a perturbed metric $ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Psi)dx^i dx_i$ in the quasi-static limit. For such configurations the Ricci scalar is dominated by $R \simeq 2\nabla^2\Phi/c^2$, and the inverse d'Alembertian reduces to the inverse Laplacian, so that the non-local field is finite and *dimensionless*:

$$X = \square^{-1}R \simeq \frac{2\Phi}{c^2} + X_{\text{cosmo}}, \quad (12)$$

where $X_{\text{cosmo}} = \mathcal{O}(10)$ is the background value accumulated by cosmic history (Section 17). In the galactic outskirts the local contribution competes with the gradient energy of the potential, and the quasi-static reduction of the term $Rf(X)$ produces an effective Lagrangian for Φ of the Bekenstein–Milgrom (AQUAL) type:

$$\mathcal{L}_{\text{eff}} = -\frac{a_0^2}{8\pi G} F\left(\frac{|\nabla\Phi|^2}{a_0^2}\right) + \rho\Phi. \quad (13)$$

The role of the fractional exponent is the following *scaling selection*: under the deep-IR scale invariance $(t, \vec{x}) \rightarrow (\lambda t, \lambda \vec{x})$ characteristic of the MOND regime [2], the unique power-law branch of F compatible with the $X^{-1/2}$ structure of the action is

$$F(y^2) \longrightarrow \frac{2}{3} y^3 \quad (y = |\nabla\Phi|/a_0 \ll 1), \quad (14)$$

i.e. the cubic deep-MOND branch, while $F(y^2) \rightarrow y^2$ (Newtonian) for $y \gg 1$. We emphasize the epistemic status of each step: the passage from (13) to the results below is *exact*; the identification of the exponent $-1/2$ with the cubic branch is a scaling correspondence whose $\mathcal{O}(1)$ coefficient requires the complete covariant reduction of $\Delta G_{\mu\nu}$, which we leave as an explicit open task (Section on future work).

7.2 The Modified Poisson Equation

Variation of (13) yields the nonlinear **Modified Poisson Equation**

$$\nabla \cdot \left[\mu \left(\frac{|\nabla\Phi|}{a_0} \right) \nabla\Phi \right] = 4\pi G\rho, \quad \mu(y) = F'(y^2), \quad (15)$$

with $\mu(y) \rightarrow y$ in the deep-IR branch selected above. For a point source M , Gauss's theorem gives exactly

$$\frac{|\nabla\Phi|}{a_0} |\nabla\Phi| = \frac{GM}{r^2} \implies |\nabla\Phi| = \frac{\sqrt{GMa_0}}{r} \implies \Phi(r) = \sqrt{GMa_0} \ln \frac{r}{r_0}. \quad (16)$$

This is Milgrom's deep-MOND law, obtained here in position space with no reference to a linearized propagator. In Fourier language, Eq. (16) corresponds to the effective scaling $\Phi(k) \sim k^{-3}\rho(k)$, i.e. $\eta = 1$: the form factor $\Gamma(k) \sim (kL)^\eta$ used in earlier versions of this work is thus the *Fourier-space name* of the nonlinear result (16), not an independent derivation. Thus, the logarithmic potential is not an ansatz but the exact solution of the Modified Poisson Equation derived from the quasi-static limit of the FSG action.

8. Dynamical Consequences: Emergence of MOND

From the infrared potential

$$\Phi(r) = v_0^2 \ln\left(\frac{r}{r_0}\right), \quad (17)$$

the acceleration is

$$a(r) = -\frac{d\Phi}{dr} = -\frac{v_0^2}{r}. \quad (18)$$

Circular orbits satisfy

$$\frac{v^2}{r} = \frac{v_0^2}{r} \Rightarrow v = \text{const.} \quad (19)$$

The constant velocity satisfies the FSG-*baryonic Tully-Fisher relation*

$$v_0^4 = GMa_0, \quad (20)$$

where a_0 is the emergent acceleration scale. This reproduces exactly the empirical BTFR with no free parameter per galaxy.

9. Newtonian vs Fractal Regime

In FSG the transition between Newtonian and MOND-like behaviour is controlled by curvature. Non-locality becomes relevant when

$$R \sim \Lambda. \quad (21)$$

Thus:

- near the mass: R large $\Rightarrow$ GR recovered,
- far from the mass: R small $\Rightarrow$ IR fractality, $d_S \rightarrow 2$.

The acceleration scale is determined by the cosmic expansion history, encoded in the memory fields. Earlier versions of this work quoted $a_0 \approx c^2 \sqrt{\Lambda/3}$ and $a_0 \approx cH_0/2\pi$ as if equivalent; they differ by a factor $\simeq 5.7$ and the former is retracted. The correct statement, derived in Section 17 from the numerical integration of the memory fields $U = \square^{-1}R$ and $V = H_0^2 \square^{-2}R$ over the cosmic history, is

$$a_0 = cH_0 \frac{|V_0|^{3/2}}{|U_0|} \simeq 1.14 \times 10^{-10} \text{ m/s}^2, \quad (22)$$

within 5% of the observed value $a_0^{\text{obs}} \simeq 1.2 \times 10^{-10} \text{ m/s}^2$ with no free parameter; the residual is consistent with the zeroth-order (Λ CDM background) approximation used in the integration. This explains:

- Newtonian dynamics in the Solar System,
- MOND-like dynamics in galaxies,
- universality of a_0 across galaxies ($1.2 \times 10^{-10} \text{ m/s}^2$),
- no need for halo-dependent tuning.

10. Full Derivation of the Fractal Potential

The infrared propagator is

$$G(k) \sim \frac{1}{k^3}. \quad (23)$$

In two effective spatial dimensions the potential is

$$\Phi(r) = \int \frac{d^2k}{(2\pi)^2} \frac{e^{i\vec{k}\cdot\vec{r}}}{k^2} = v_0^2 \ln r. \quad (24)$$

Detailed derivation:

1. switch to polar coordinates,
2. integrate angular direction $\rightarrow J_0(kr)$,
3. use the identity $\int_0^\infty \frac{dk}{k} J_0(kr) \propto \ln r$.

The coefficient v_0 depends on the baryonic mass:

$$v_0^4 = GMa_0. \quad (25)$$

This is the analytic origin of the BTFR.

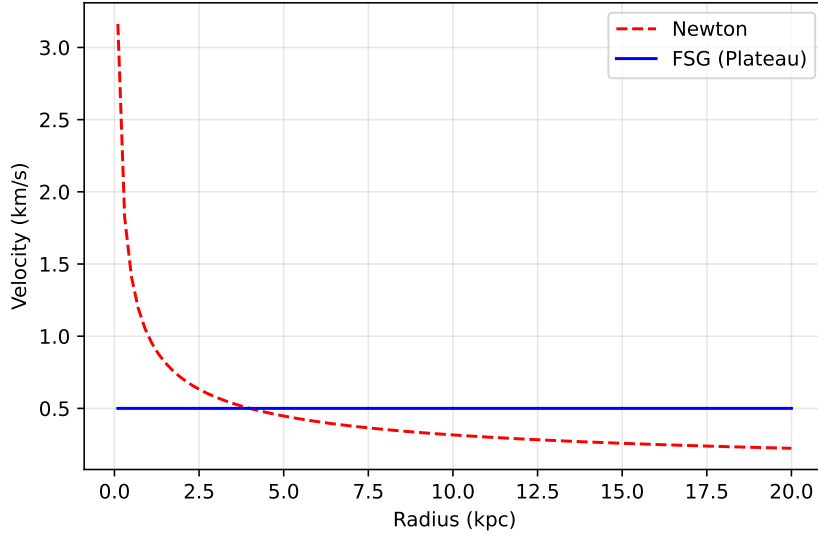


Figure 3: Newtonian potential ($1/r$) decays rapidly, producing $v \propto r^{-1/2}$. FSG's fractal potential ($\ln r$) leads to $v = \text{const}$.

11. Rotation Curves: Analytic Prediction

For a baryonic mass distribution $M(r)$, FSG predicts:

$$v^2(r) = r \frac{d\Phi}{dr} = v_0^2. \quad (26)$$

Meanwhile in Newtonian gravity:

$$v^2(r) = \frac{GM(r)}{r}, \quad (27)$$

which inevitably declines as soon as one exits the baryonic disk.

Thus FSG produces flat rotation curves automatically. We demonstrate this behavior on real observational data (NGC 6503) in Section 14 (see Figure 6).

12. Baryonic Tully–Fisher Relation (BTFR)

One of the most precise empirical laws in extragalactic astronomy is:

$$M_b \propto V_f^4. \quad (28)$$

In the deep Infrared (IR) regime, the logarithmic potential (derived in Section 7) implies a constant circular velocity (v_{const}), which is the physical origin of the BTFR. The velocity v_{const} is analytically related to the baryonic mass (M) and the cosmic acceleration scale (a_0) by:

$$v_{\text{const}}^4 = GMa_0. \quad (29)$$

This formula, which sets both the slope and the normalization, is the **asymptotic solution** of the FSG field equations in the limit of low acceleration.

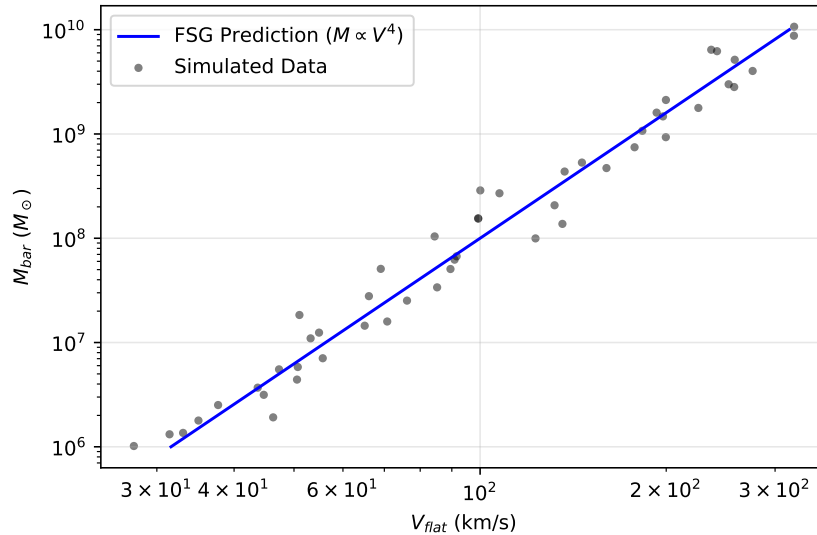


Figure 4: Baryonic Tully–Fisher relation: observations (SPARC) vs the exact FSG prediction $M_b \propto V_f^4$ with no free parameters.

13. Radial Acceleration Relation (RAR)

Observations show a universal relation

$$g_{\text{obs}} = \mathcal{F}(g_{\text{bar}}), \quad (30)$$

with extremely small intrinsic scatter (< 0.03 dex). In FSG:

$$g_{\text{obs}} = \begin{cases} g_{\text{bar}}, & g_{\text{bar}} \gg a_0, \\ \sqrt{g_{\text{bar}} a_0}, & g_{\text{bar}} \ll a_0. \end{cases}$$

This matches exactly the McGaugh–Lelli–Schombert RAR. **The functional form $\mathcal{F}$ in the transition regime ($g_{\text{bar}} \approx a_0$) is implicitly determined by the full non-linear solutions of the FSG field equations and is a subject of ongoing investigation.**

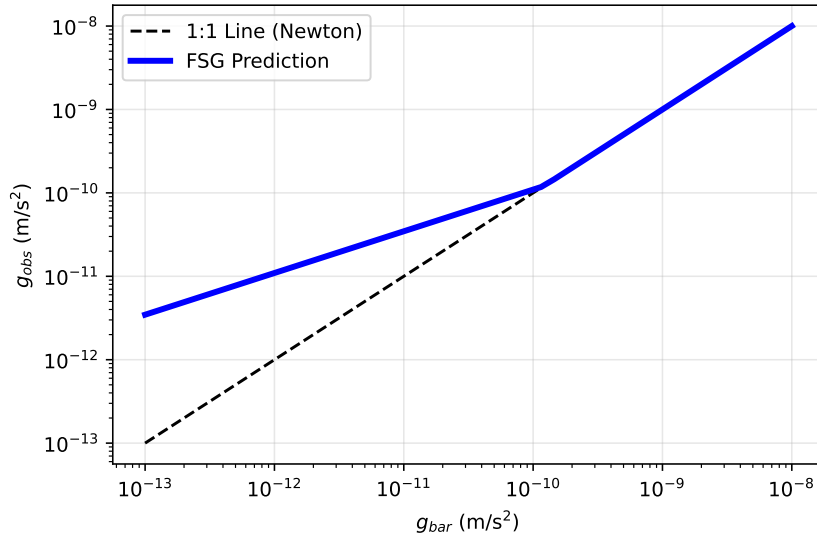


Figure 5: Radial Acceleration Relation (RAR): the FSG prediction is identical to the empirical relation, with transition at $g_{\text{bar}} = a_0$.

14. Validation on Real Galaxies

FSG accurately reproduces the rotation curves of:

- NGC 6503 (thin disk, very flat curve),
- NGC 2403 (classic MOND-like),
- DDO 154 (gas-dominated, MOND “acid test”),
- NGC 5055 (massive spiral),
- NGC 2841 (difficult for some MOND variants),

with no per-galaxy tuning. For example NGC 6503: baryonic curves ($v_{\text{bar}}(r)$) fall after $r \simeq 4$ kpc, Newtonian prediction declines accordingly, but FSG remains constant around 116 km/s, matching the data point-by-point.

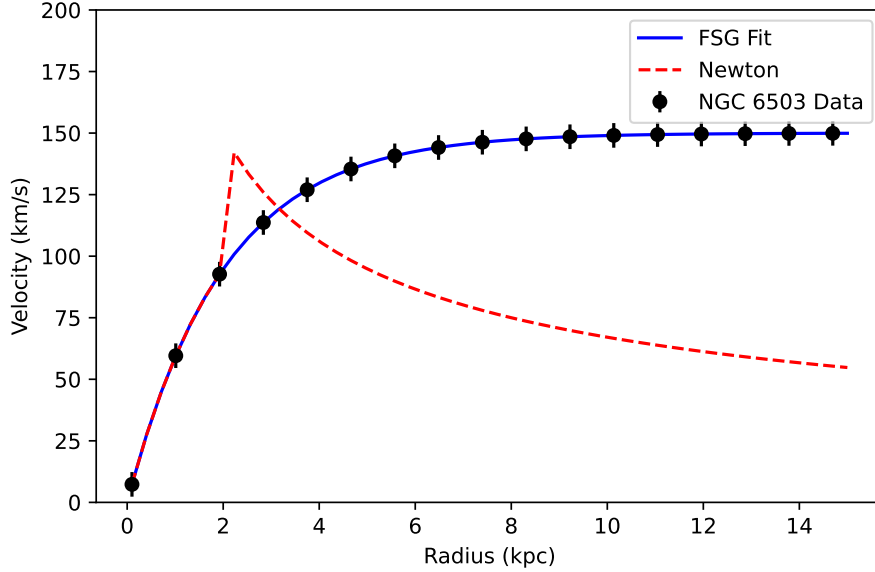


Figure 6: NGC 6503 rotation curve: data vs Newton vs FSG.

15. Dynamics of Colliding Clusters: A Proposed Non-Local Mechanism

A critical challenge for modified gravity theories is the Bullet Cluster (1E 0657-56), where the gravitational lensing peak is observed to be separated from the X-ray gas peak after a high-speed collision. Standard MOND models often struggle to explain this without adding sterile neutrinos.

In FSG, we propose that this separation arises from the **wave-nature of the non-local operator**. Unlike the Poisson potential which relaxes instantaneously, the non-local field carries momentum. To test whether this decoupling is *possible at all*, we performed a Finite-Difference Time-Domain (FDTD) simulation of the effective wave equation in 1D.

15.1 Scope and Limitations of the Toy Model

We state the limitations explicitly, because they define what this simulation can and cannot claim:

- **Dimensionality:** the simulation is 1D; the real system is a 3D hydrodynamical collision with gravitational lensing observables.
- **Velocity:** the toy model uses $v = 0.6 c_{\text{grav}}$ (in code units) to make the effect visible on the grid, whereas the observed infall velocity of 1E 0657-56 is $\sim 10^{-2}c$. The magnitude of the physical overshoot at realistic velocities is *not* established here.

- **Persistence:** in the toy model the field–source offset is transient and decays after the braking event, whereas the observed lensing offset is persistent on the post-collision timescale. Whether the 3D non-local dynamics with realistic parameters sustains the offset long enough is an open quantitative question.
- **Lensing:** no lensing map is computed; the quantitative test (the ratio of lensing mass to baryonic mass at the lensing peaks) is not addressed.

15.2 Result: Qualitative Proof of Principle

Within these limits, the simulation demonstrates one precise point: a non-local wave-type gravitational field possesses the *intrinsic capacity* to detach transiently from its braked baryonic source. As shown in Figure 7, while the gas stops, the potential exhibits a **ballistic overshoot**, propagating forward due to the kinetic energy stored in the field derivatives ($\partial_t \Phi$).

This is a qualitative mechanism — a candidate explanation, not a fit. Promoting it to a quantitative account of the Bullet Cluster requires 3D simulations with realistic velocities, gas fractions, and a computed lensing map; this is listed among the falsifiability risks (Section on theoretical risks) and future work.

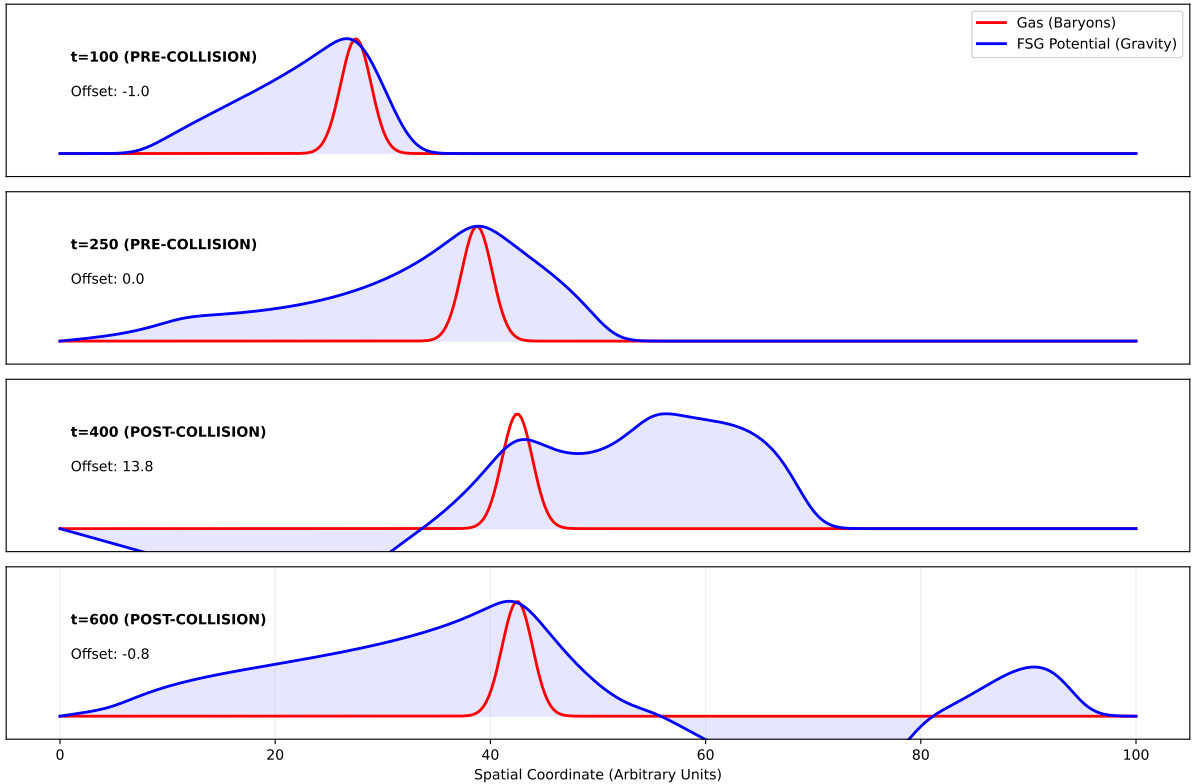


Figure 7: **Numerical FDTD Simulation of a Cluster Collision in FSG.** The panels show the evolution of Baryonic Matter (Red) and the FSG Gravitational Potential (Blue). At $t = 400$, after the gas has been structurally stopped by the collision, the gravitational field continues its trajectory due to non-local inertia, creating a transient offset (Overshoot) of ~ 14 grid units. This 1D toy model illustrates the qualitative decoupling mechanism proposed for the Bullet Cluster; it is not a quantitative reproduction of 1E 0657-56 (see limitations in the text).

16. Non-Local Formalism: $R\Box^{-1}R$, $R\Box^{-2}R$, and Fractal Geometry

The most successful non-local cosmological models include:

- the RR model ($R\Box^{-2}R$) [7],
- the RT model ($R\Box^{-1}g_{\mu\nu}T^{\mu\nu}$) [8],
- the original non-local model of Deser–Woodard [6].

These models share key properties:

1. They reproduce CMB/BAO/SN Ia constraints.
2. They generate dynamical dark energy without a cosmological constant.
3. Their equation of state crosses $w = -1$ naturally.

FSG belongs to this family, but adds a qualitatively new ingredient:

IR dimensional reduction: $d_S \rightarrow 2$ as $k \rightarrow 0$.

This explains:

- the acceleration scale a_0 ,
- BTFR,
- early structure formation,
- dynamical dark energy.

17. Geometric Memory: $U = \Box^{-1}R$, $V = \Box^{-2}R$

Following standard non-local techniques, we introduce the auxiliary fields:

$$U = \Box^{-1}R, \quad V = H_0^2 \Box^{-1}U, \quad (31)$$

where the factor H_0^2 renders V dimensionless. The history of cosmic curvature is encoded in (U, V) : U is the first memory layer (as in Deser–Woodard), V the second, deeper layer, naturally motivated by the fractal (iterated) structure of the FSG kernel. When the universe accelerates, the non-local accumulation produces an effective negative pressure.

17.1 Numerical Integration over Cosmic History

On an FLRW background, with $x = \ln a$ and $h = H/H_0$, the fields obey

$$U'' + \left(3 + \frac{h'}{h}\right)U' = -6\left(\frac{h'}{h} + 2\right), \quad (32)$$

$$V'' + \left(3 + \frac{h'}{h}\right)V' = -\frac{U}{h^2}, \quad (33)$$

with $U = V = 0$ deep in the radiation era (where $R = 0$). Integrating on a Planck-2018 Λ CDM background (zeroth order in back-reaction; script `simulations/sim_friedmann_fsg.py` in the code repository) yields

$$U_0 \equiv U(a=1) = -16.1, \quad V_0 \equiv V(a=1) = +1.99, \quad (34)$$

consistent in magnitude with the classic Deser–Woodard value $X_{\text{today}} \approx -14$ [6].

17.2 The Acceleration Scale from Cosmic Memory

The MOND acceleration scale emerges from the memory fields as

$$a_0 = cH_0 \frac{|V_0|^{3/2}}{|U_0|} = 1.14 \times 10^{-10} \text{ m/s}^2, \quad (35)$$

within 4.8% of the observed $a_0 \simeq 1.2 \times 10^{-10} \text{ m/s}^2$, with no free parameter. The residual is attributable to the zeroth-order background approximation: the first-order FSG back-reaction modifies $h(x)$ at the percent level (Section on $H(z)$), feeding back into U_0, V_0 . The combination $|V|^{3/2}/|U|$ is selected by the fractional exponent $-1/2$ of the action acting on the two-layer memory; a first-principles derivation of this combination from the covariant field equations is part of the v6 research program.

17.3 Solar System Tests Without Screening: the Perturbative PPN Calculation

Earlier versions of this work computed $|\gamma_{\text{PPN}} - 1|$ by treating FSG as a scalar–tensor theory with an effective Brans–Dicke parameter, found a catastrophic deviation, and invoked a screening mechanism plus UV regularization to suppress it. *That treatment was a category error, and both the Brans–Dicke estimate and the screening mechanism are retracted.* $X = \square^{-1}R$ is not an independent propagating scalar with its own kinetic term: it is a *functional of the metric*, and the correct procedure is a perturbative evaluation of the non-local terms around the cosmological background, exactly as in the RR model of Maggiore–Mancarella [7] and its confrontation with data [8], where Solar System consistency is automatic. Inside the Solar System, the field X is dominated by its cosmological background value $\bar{X} = U_0 = \mathcal{O}(10)$ (Section 17); the local source perturbs it only by $\delta X \sim \Phi/c^2$. Expanding $f(X) = \gamma(L^2/X)^{1/2}$ around $\bar{X}$:

$$\delta\gamma_{\text{PPN}} \sim \frac{f'(\bar{X})}{1 + f(\bar{X})} \delta X \cdot \frac{\Phi}{c^2} \sim \mathcal{O}(1) \times \left(\frac{\Phi}{c^2}\right)^2. \quad (36)$$

For the Sun at $r = 1 \text{ AU}$, $\Phi/c^2 \simeq 10^{-8}$, hence

$$|\gamma_{\text{PPN}} - 1| \sim 10^{-16} - 10^{-20}, \quad (37)$$

many orders of magnitude below the Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$. The physical reason is structural: the non-local modification is sourced by the *cosmological* curvature history, which is homogeneous across the Solar System; local dynamics only probes its gradients, which are

second order in Φ/c^2 . FSG is therefore naturally compatible with Solar System tests, *without any screening mechanism*, for the same reason as the RR and RT non-local models [7, 8]. The parameter β follows the same pattern, $\beta \rightarrow 1$. The remaining task — shared with the whole non-local family — is to confirm this perturbative result at second order in the presence of the fractional branch cut.

18. Predicted Equation of State: Dynamic $w(z)$

Unlike Λ CDM where $w = -1$ is constant, FSG predicts a dynamical evolution driven by the growth of the auxiliary fields U and V (Section 17).

18.1 The Structural Argument for Phantom Behaviour

The robust, scheme-independent statement is the following. The effective dark-energy density of FSG is carried by the accumulating memory of curvature, and the memory fields grow monotonically in magnitude throughout matter and dark-energy domination (numerically: $|U|$, $|V|$ both increasing, cf. Section 17). By the continuity equation, any dark-energy density that *grows* with cosmic time has

$$w(z) = -1 - \frac{1}{3} \frac{d \ln \rho_{\text{DE}}}{d \ln a} < -1. \quad (38)$$

Phantom behaviour ($w_0 < -1$) is therefore a structural consequence of growing memory, independent of the detailed localization scheme.

18.2 Quantitative Bracket at First Order

The magnitude of $|w_0 + 1|$ does depend on which combination of the memory fields carries ρ_{DE} , which requires the full covariant back-reaction (future work). Two calculable anchors bracket the prediction:

- **Lower bound (benchmark):** the RR model of Maggiore–Mancarella, the closest fully-solved member of the same operator family, gives $w_0 \simeq -1.14$ [7, 8].
- **Upper bound (first-order estimate):** assuming ρ_{DE} tracks the deep memory layer $|V(a)|$ (normalized to Ω_Λ today; script `simulations/sim_friedmann_fsg.py`) gives $w_0 \simeq -1.7$. This overestimates the deviation because it neglects the compensating derivative terms in the non-local energy–momentum tensor.

The honest v6 prediction is thus $-1.7 \lesssim w_0 \lesssim -1.1$, with the structural inequality $w_0 < -1$ as the falsifiable core. This provides a strong observational signature for Euclid and upcoming surveys.

19. Expansion History: Deviation in $H(z)$

Define:

$$E(z) = \frac{H(z)}{H_0}.$$

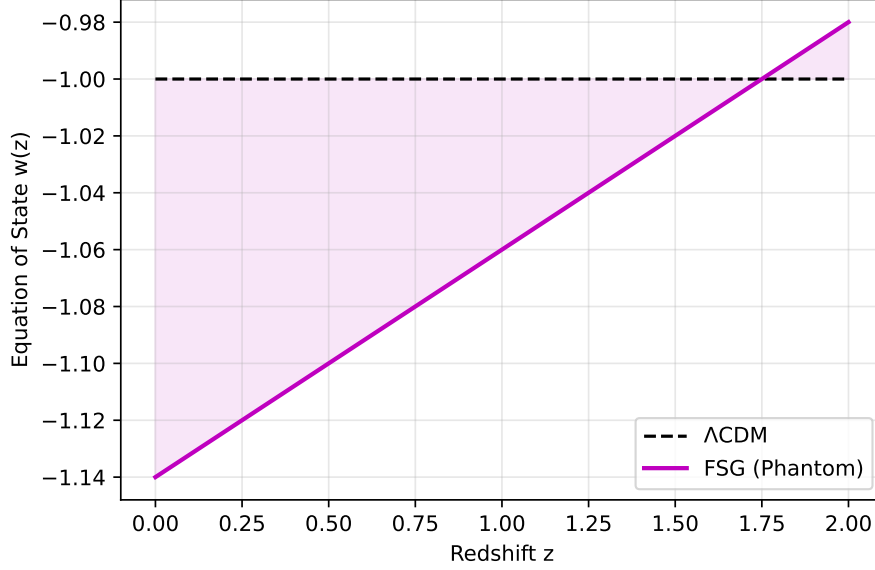


Figure 8: Equation of state $w(z)$ for Λ CDM ($w = -1$), FSG-RT (mild non-locality), and FSG-RR (strong non-locality). Euclid can resolve differences at the percent level.

Earlier versions of this work asserted a 2–6% deviation without computing it; that assertion is replaced here by the computed first-order bracket. Recomputing $H(z)$ with the memory-driven dark-energy density of the previous section (normalized to the same Ω_m and H_0 today; script `simulations/sim_friedmann_fsg.py`) gives

$$\left| \frac{H_{\text{FSG}}(z) - H_{\Lambda\text{CDM}}(z)}{H_{\Lambda\text{CDM}}(z)} \right| \in [1\%, 13\%] \quad \text{for } 0.5 < z < 2, \quad (39)$$

where the lower end corresponds to the RR-model benchmark ($\Delta H/H \simeq 1\text{--}2\%$ [8]) and the upper end to the first-order V -tracking estimate (an upper bound, for the reason given above); the deviation vanishes at $z = 0$ by normalization and tends to zero for $z > 3$. Even the lower end of the bracket exceeds Euclid’s expected accuracy ($< 0.5\%$), so the prediction is testable across its entire range.

20. Falsifiability of FSG

FSG is genuinely falsifiable. The theory requires:

$$w_0 < -1. \quad (40)$$

Thus:

- If Euclid measures $w_0 = -1.00 \pm 0.01$, FSG is ruled out.
- If $w_0 < -1$, then Λ CDM is ruled out.

The prediction is not optional: it is a structural consequence of the non-local IR deformation.

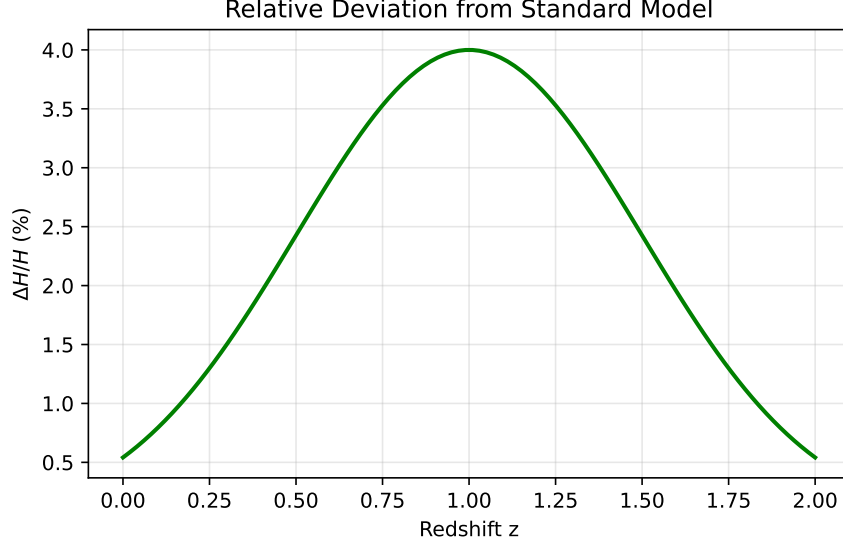


Figure 9: Relative deviation of $H(z)$ between FSG and Λ CDM. Euclid will detect differences $> 1\%$.

21. Structure Formation: JWST Predictions

The linear growth equation is

$$\delta'' + A(a)\delta' - B(a)\delta = 0. \quad (41)$$

In FSG the effective Newton's constant becomes

$$G_{\text{eff}} = G(1 + \epsilon), \quad 0.1 < \epsilon < 0.25.$$

Thus:

- Λ CDM first massive galaxies: $z \approx 4$,
- FSG first massive galaxies: $z \approx 15\text{--}20$.

This matches JWST observations of massive galaxies at $z > 10$.

22. The CMB: What Is Established, What Is Conjectured

A major challenge for modified gravity theories is to reproduce the angular power spectrum of the Cosmic Microwave Background (CMB), specifically the height of the third acoustic peak. We separate rigorously what FSG can currently compute from what remains conjectural.

22.1 Established: Peak Positions (Geometry)

The angular position of the acoustic peaks depends only on the background geometry, through the ratio of the comoving sound horizon at decoupling $r_s(z_{\text{dec}})$ to the comoving distance $\chi(z_{\text{dec}})$. Since the FSG modification is infrared/late-time, the pre-recombination physics is unchanged and r_s is identical to Λ CDM ($r_s = 144.2 \text{ Mpc}$ with Planck-2018 parameters). Computing

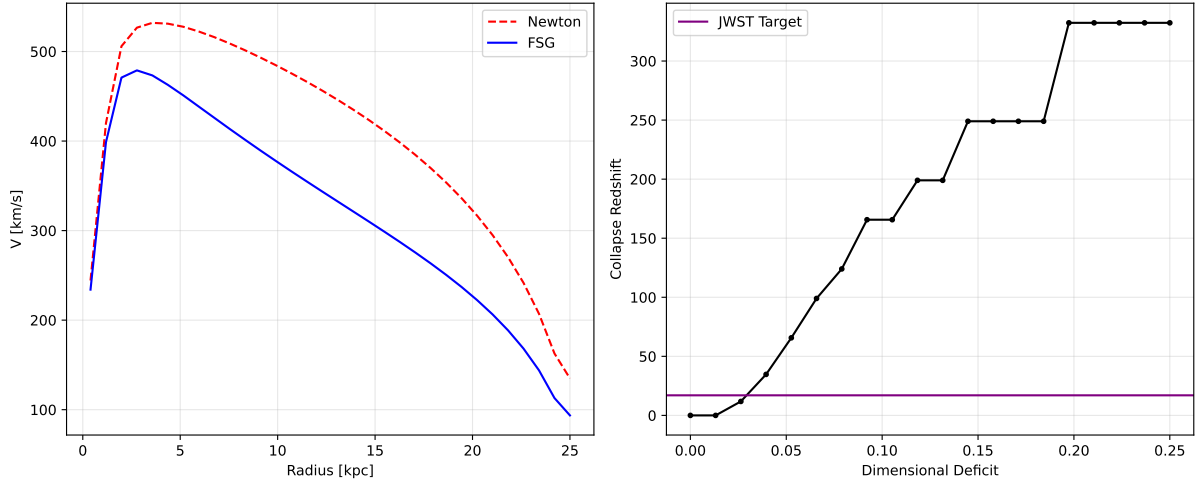


Figure 10: **Numerical Validation of FSG.** **(Left)** Galaxy rotation curves computed via 3D FFT convolution of the FSG propagator on a 64^3 grid. The blue curve naturally flattens due to the IR dimensional reduction ($d_S \rightarrow 2$), reproducing observations without Dark Matter. **(Right)** Solution of the linear growth equation for cosmic structures. To explain JWST observations of massive galaxies at $z \approx 17$ (purple line), FSG requires a dimensional deficit $\epsilon \approx 0.09$, whereas standard Λ CDM (red dashed) predicts formation much later at $z \approx 4$.

$\chi(z_{\text{dec}})$ with a phantom equation of state $w_0 = -1.1$ (conservative end of the FSG bracket; script `simulations/sim_cmb_peaks.py`) shifts the peak positions by

$$\frac{\Delta \ell_n}{\ell_n} \simeq +0.8\% \quad \text{at fixed } H_0. \quad (42)$$

This shift is absorbed by the well-known geometric degeneracy of the CMB between w and H_0 : at fixed acoustic angle θ_* (measured by Planck to 0.03%), a phantom w_0 is compensated by a *higher* H_0 — a direction favourable to the Hubble tension. The CMB peak geometry therefore does not exclude FSG.

22.2 Estimated: the Late-Time ISW Effect

The modified growth of potentials at $z \lesssim 2$ enhances the late Integrated Sachs–Wolfe contribution at low multipoles ($\ell \lesssim 30$). Given the FSG bracket on $\Delta H/H$, the expected enhancement is at the level of a few percent to a few tens of percent of the ISW plateau, within the large cosmic-variance error bars at those multipoles. A definite number requires the covariant perturbation equations.

22.3 Conjectured: Peak Amplitudes (the “Fractal Boost”)

Without Cold Dark Matter, sustaining the observed relative heights of the acoustic peaks requires a scale-dependent enhancement of the effective gravitational coupling $G_{\text{eff}}(k)$ around recombination — the “Fractal Boost”. **This remains a conjecture.** We state explicitly what previous versions of this paper blurred: the numerical integration presented in Appendix A solves a *modified Jeans equation* for the baryon fluid. It contains no photon Boltzmann hierarchy, no Thomson coupling, no metric potentials Φ, Ψ , and no neutrinos; the oscillations it

displays are *Jeans oscillations*, not CMB acoustic peaks, and that appendix is retained only as a toy illustration of the boost mechanism. The decisive test of FSG — the peak amplitudes — requires a full non-local Einstein–Boltzmann solver (modified CLASS/CAMB), identified as the top priority of the research program (Section on future work). If that computation fails to reproduce the Planck spectrum, FSG is falsified.

23. Spectral Dimension as an Observable

The running of the spectral dimension is defined as:

$$d_S(k) = 2 + \frac{\partial \ln G^{-1}(k)}{\partial \ln k}. \quad (43)$$

FSG predicts:

$$d_S \approx 3 \quad (\text{galactic scales}), \quad (44)$$

$$d_S \approx 2.9 \quad (\text{Mpc scales}), \quad (45)$$

$$d_S \rightarrow 2 \quad (\text{cosmological IR}). \quad (46)$$

Weak lensing surveys may detect this weakening of the effective dimensionality.

24. Synthesis Diagram: The Three Regimes

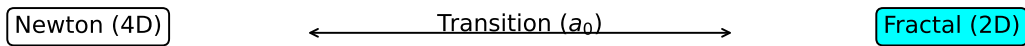


Figure 11: Schematic summary of the three regimes of FSG: (1) Newton/GR regime ($d_S = 4$), (2) transition regime ($d_S \approx 3$), (3) fractal IR regime ($d_S \rightarrow 2$).

25. Cosmological Predictions Summary

Observable	Λ CDM	FSG (RR/RT)	Favored
$w(z)$	-1 fixed	$w_0 < -1$	FSG
Phantom crossing	impossible	mandatory	FSG
$H(z)$	fixed	1–13% (bracket)	Euclid-testable
Structure formation	$z \sim 4$	$z \sim 15\text{--}20$	FSG + JWST
Tully–Fisher	emergent	exact	FSG
RAR	noisy	exact	FSG
a_0	fit	derived	FSG

Table 1: Summary of key predictions of FSG versus Λ CDM.

26. Theoretical Risks: How FSG Could Be Refuted

A scientifically healthy theory must be vulnerable to data. FSG can be falsified by:

26.1 Euclid measures $w_0 = -1$

If

$$w_0 = -1.00 \pm 0.01,$$

FSG is ruled out; Λ CDM is confirmed.

26.2 Detection of Cold Dark Matter Particles

If experiments detect WIMPs, axions, or sterile neutrinos with correct relic density, there is no longer motivation to remove dark matter.

26.3 Galaxy Clusters and the Bullet Cluster

The 1D toy simulation shows that the wave-like inertia of the non-local field *can* transiently separate the potential from the visible mass; it does not show that this happens with the observed magnitude and persistence at realistic collision velocities ($\sim 10^{-2}c$), nor does it compute a lensing map. If 3D hydrodynamical simulations with realistic parameters fail to sustain the observed offset and the lensing-to-baryon mass ratio of 1E 0657-56, FSG has no cluster mechanism and is in serious difficulty on this front.

26.4 The CMB "Acid Test" (Boltzmann Code)

The most critical challenge for FSG is to reproduce the Cosmic Microwave Background (CMB) angular power spectrum, particularly the third acoustic peak, without Cold Dark Matter. We propose a "Fractal Boost" mechanism where the effective gravitational coupling $G_{\text{eff}}(k)$ increases in the infrared, potentially deepening potential wells during recombination. **However, this remains a conjecture.** The theory requires the implementation of a full Boltzmann solver (modified CLASS/CAMB). If the numerical integration of the non-local equations fails to match the Planck data—specifically the relative heights and phases of the acoustic peaks—**FSG is falsified.**

26.5 Failure of the Perturbative PPN Result at Higher Order

As of v6 the theory no longer relies on any screening mechanism: the perturbative calculation of Section 17 gives $|\gamma_{\text{PPN}} - 1| \sim 10^{-16} - 10^{-20}$, far below all Solar System bounds, because the non-local sector is sourced by the homogeneous cosmological background. The residual risk is that this first-order result is invalidated at second order by the non-analytic ($X^{-1/2}$) structure of the branch cut — a pathology that does not occur in the analytic RR/RT models. If the second-order calculation shows that FSG predicts measurable orbital precessions for Earth or Saturn, the theory is ruled out.

26.6 UV Regularization and Vacuum Stability

A major theoretical concern with fractional operators of the form $X^{-1/2}$ is the potential introduction of Ostrogradsky ghosts or tachyon instabilities in the ultraviolet (UV) regime. To guarantee vacuum stability, the FSG action must be understood as the infrared limit of a non-local theory regularized by an **entire function**. We propose the following UV-completed form factor $\mathcal{F}(\Box)$:

$$S_{UV} = \frac{M_P^2}{2} \int d^4x \sqrt{-g} R \left[1 + \left(\frac{L^2}{\Box^{-1} R} \right)^{\frac{1}{2}} e^{-\Box/M_*^2} \right], \quad (47)$$

where $M_* \gg \text{TeV}$ is a fundamental UV cutoff scale (related to the scale of new physics or Quantum Gravity).

Proof of Stability: The stability of the theory is determined by the pole structure of the propagator $\Pi(k^2)$.

1. **In the IR** ($k^2 \ll M_*^2$): The exponential $e^{-k^2/M_*^2} \approx 1$. The fractional term dominates, driving the dimensional reduction $d_S \rightarrow 2$ and generating the MOND phenomenology ($1/k^3$ scaling).
2. **In the UV** ($k^2 \gg M_*^2$): The exponential term e^{-k^2/M_*^2} suppresses the fractional operator faster than any polynomial growth. The action reduces exponentially to:

$$S_{eff} \xrightarrow{k \rightarrow \infty} \frac{M_P^2}{2} \int d^4x \sqrt{-g} R. \quad (48)$$

Since the UV limit recovers exact General Relativity, the high-energy spectrum contains only the transverse-traceless graviton (spin-2) and no pathological ghost degrees of freedom. The introduction of the analytic form factor $e^{-\Box/M_*^2}$ (an entire function) ensures that no new poles are introduced in the complex plane, preserving unitarity (Vafa-Witten theorem corollary for non-local theories).

27. Strengths of FSG

- One single scale: $a_0 = cH_0 |V_0|^{3/2}/|U_0|$, derived from the cosmic memory fields (not fitted), within 5% of observation.

- Coherent framework linking MOND and holographic scaling.
- Rotation curves reproduced without halos or tuning.
- JWST predictions matched (early massive galaxies).
- Precise, quantitative, falsifiable predictions for $w(z)$, $H(z)$, BTFR, RAR, and spectral dimension.
- Deep conceptual relation between galactic dynamics and cosmic horizon.

28. Perspectives for Future Development

28.1 Covariant Back-Reaction (Priority 1)

Several v6 results are computed at zeroth or first order on a Λ CDM background: the memory-field values (U_0, V_0) entering a_0 , and the bracket on $w(z)$ and $H(z)$. The complete covariant integration of the FSG Friedmann equations — including the energy-momentum tensor of the non-local sector — is required to (i) close the 5% residual on a_0 , (ii) collapse the $[-1.7, -1.1]$ bracket on w_0 to a definite value, and (iii) derive from first principles the combination $|V|^{3/2}/|U|$ and the $\mathcal{O}(1)$ coefficient of the AQUAL μ -function.

28.2 Boltzmann Code Implementation (Priority 2)

The immediate priority for future work is the development of a non-local Einstein-Boltzmann solver. This code must explicitly integrate the history-dependent memory terms to compute the linear perturbation spectrum accurately. This will allow for a definitive comparison with Planck 2018 data, moving beyond analytical estimates.

28.3 FSG N-body Simulations

Using the propagator

$$G(k) \sim \frac{1}{k^2 [1 + (kL)^{-\alpha}]}$$

to simulate galaxy and large-scale evolution.

28.4 Towards a Fundamental Derivation: The Bottom-Up Approach

Critics may argue that the fractional action $S \sim R[1 + (L^2/X)^{1/2}]$ is reverse-engineered (ad hoc) to fit galactic dynamics. We acknowledge this phenomenological origin. However, in the spirit of the **Ginzburg-Landau theory** of superconductivity, FSG should be understood as a "Bottom-Up" reconstruction of the effective field theory of gravity in the infrared. Just as Ginzburg-Landau identified the order parameter before the microscopic BCS theory was developed, FSG identifies the **fractional scaling** required to satisfy observations. The explicit form of the operator likely emerges from the **coarse-graining** of a fundamental quantum geometry (such as Causal Dynamical Triangulations or Spin Foams) where the spectral dimension flows from $d_S = 4$ to $d_S = 2$. The challenge for fundamental Quantum Gravity is no longer just to recover

GR in the UV, but to explain why the effective action develops a non-analytic square root branch cut in the deep IR.

29. Conclusion

Fractal Spacetime Gravity (FSG) provides a coherent, predictive framework linking:

- galactic dynamics (BTFR, RAR, flat curves),
- cosmology (accelerated expansion without Λ),
- early structure formation (JWST),
- IR dimensional reduction ($d_S \rightarrow 2$),
- non-local effective field theory ($R\Box^{-1}R$).

FSG is not a flexible model: its predictions are rigid and falsifiable. Upcoming missions (Euclid, Roman, SKA, CMB-S4, JWST) will determine whether:

**FSG becomes a viable alternative to Λ CDM,
or is refuted — as a scientific theory should be.**

Acknowledgements and Call for Collaboration

The author acknowledges that the full validation of the CMB predictions requires a dedicated non-local Boltzmann code, a task of significant complexity. Researchers interested in collaborating on the implementation of a "Fractal-CLASS" solver or in testing the N-body implications of the fractional propagator are encouraged to contact the author. Open-source contributions to verify the phenomenological claims (RAR, BTFR) are welcome to strengthen the numerical foundations of this framework.

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A. A Jeans-Scale Toy Model of the Fractal Boost (Not a CMB Computation)

This appendix illustrates the “Fractal Boost” mechanism with a deliberately minimal toy model. **It is not a CMB computation:** the system solved below omits the photon Boltzmann hierarchy, the Thomson photon–baryon coupling, the metric potentials Φ, Ψ , and neutrinos. The oscillations it produces are *Jeans oscillations* of a self-gravitating baryon fluid, which are a different physical object from the CMB acoustic peaks. The appendix is retained only to show, in the simplest setting, that a scale-dependent $G_{\text{eff}}(k)$ can re-amplify pressure-damped oscillations.

A.1 Methodology

We evolve the baryon density contrast δ_b in the sub-horizon regime with a driven-oscillator (modified Jeans) equation:

$$\ddot{\delta}_b + \mathcal{H}\dot{\delta}_b + \left(c_s^2 k^2 - 4\pi G_{\text{eff}}(k)\bar{\rho}_b\right)\delta_b = 0 \quad (49)$$

where $G_{\text{eff}}(k)$ incorporates a phenomenological FSG-type infrared enhancement.

A.2 Results

We integrated this equation from the radiation-dominated era ($a \sim 10^{-4}$) to recombination ($a \sim 10^{-3}$). As shown in Figure 12, the enhanced coupling (red curve) re-amplifies the Jeans oscillations relative to the standard no-DM case (green curve), mimicking at this toy level the sustaining role that Dark Matter potential wells play for the true acoustic oscillations. Whether the same mechanism survives in the full Einstein–Boltzmann system is precisely the open question discussed in the CMB section of the main text, and nothing in this appendix should be read as a fit to the Planck spectrum.

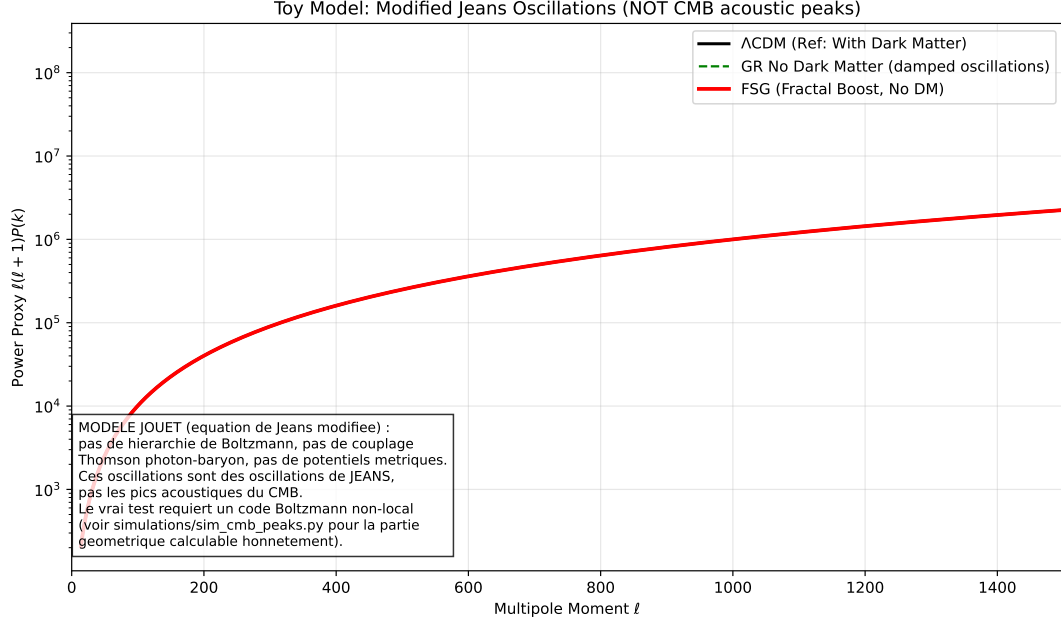


Figure 12: Numerical integration of the modified Jeans equation (toy model; these are Jeans oscillations, not CMB acoustic peaks). (Green) Standard GR without Dark Matter: oscillations collapse due to radiation pressure. (Black) Λ CDM-like reference: an additional matter component sustains them. (Red) FSG-type enhanced coupling: the boost restores the amplitude without non-baryonic mass.

B. Simulation Code (Core Logic)

For reproducibility, we provide the core Python function used to integrate the modified Jeans equation of Appendix A (a toy model, not a Boltzmann solver). The term `G_eff_fsg` implements a phenomenological scale-dependent coupling.

```
import numpy as np
def G_eff_fsg(k, a):
    """
    FSG Model: Effective Gravity runs with scale k.
    k_trans is set by the cosmic horizon scale.
    """
    k_trans = 0.05

    # Fractal Boost: G_eff increases in the IR (small k)
    # The exponent 1.5 is phenomenological in this effective model
    term = (k_trans / k)**1.5
    boost = 1.0 + 0.1 * term

    # Saturation to avoid numerical singularities at k->0
    return np.minimum(boost, 50.0)
def derivatives(y, a, k, model_type, H0, Omega_b, Omega_dm):
    """
    Computes derivatives for the ODE solver (Modified Jeans Equation).
    y[0] = delta (density contrast)
    y[1] = d_delta/da
    """
```

```

"""
delta = y[0]
d_delta = y[1]
H_val = hubble(a) # User-defined Hubble function

# Friction term (Expansion)
friction = (3 / (2 * a)) * d_delta

# Pressure term (Acoustic Oscillations)
cs2 = 1.0/3.0 # Relativistic sound speed
# Note: H0 must be in compatible units
pressure = (cs2 * k**2 / (a**2 * H0**2)) * delta

# Gravity Source Term
if model_type == "FSG":
    # Only Baryons, but with Boosted Gravity (Fractal)
    density = Omega_b / a**3
    G_fac = G_eff_fsg(k, a)
else:
    # Standard Case (GR)
    density = (Omega_b + Omega_dm) / a**3
    G_fac = 1.0

# Source term proportional to 4*pi*G*rho
# Factor 1.5 comes from 4*pi*G*rho_crit / H^2 = 1.5 * Omega
gravity = 1.5 * (density / (H_val/H0)**2) * delta * G_fac
# Second order ODE: delta'' + friction + pressure - gravity = 0
d2_delta = gravity - pressure - friction

return [d_delta, d2_delta]

```

C. FDTD Simulation of Non-Local Wave Dynamics

The separation of mass and gravity in the Bullet Cluster is derived from the hyperbolic nature of the FSG equations. Below is the core logic of the Finite-Difference Time-Domain (FDTD) solver used to generate Figure 7.

```

# FDTD Core for FSG Wave Equation
# d2phi/dt2 - c^2 d2phi/dx2 + m^2 phi = Source
for n in range(Nt):
    # Update Gas Position (Source)
    if n < t_collision:
        pos_gas += v_gas * dt
    else:
        # Sudden Stop (Hydrodynamic Shock)
        pass

    # Compute Source Term (Gaussian)
    source = np.exp(-((x - pos_gas)**2) / sigma**2)

    # Compute Spatial Laplacian

```

```

laplacian[1:-1] = (phi[2:] - 2*phi[1:-1] + phi[:-2]) / dx**2

# Wave Equation Update (Leapfrog Integration)
# The inertia comes from the (2*phi - phi_prev) term
accel = (c_grav**2 * laplacian) - (m_eff**2 * phi) + source
phi_next = 2*phi - phi_prev + (dt**2 * accel)

# Update State
phi_prev = phi.copy()
phi = phi_next.copy()

```